

VIRTUSU, Estonia

WMO: 261280

Lat: 58.5728N

Lon: 23.5136E

Elev: 4

StdP: 101.28

Time Zone: 2.00 (EUE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.0	-14.8	-20.5	0.6	-17.4	-17.7	0.8	-14.4	11.3	-1.6	9.8	-1.4	2.0	100	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	26.2	20.1	24.3	19.2	22.8	18.3	21.1	24.9	20.0	23.5	18.9	22.1	2.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.4	23.4	18.5	13.4	22.2	17.5	12.6	21.1	61.0	24.7	57.1	23.5	53.8	22.2	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.0	7.0	DB	-20.1	28.3	5.0	2.3	-23.7	30.0	-26.6	31.3	-29.5	32.6	-33.1	34.3	(4)
(5)			WB	-20.4	22.1	4.4	1.3	-23.6	23.0	-26.2	23.8	-28.6	24.5	-31.9	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-3.1	-3.6	-1.0	4.7	11.1	15.0	18.5	17.6	13.1	7.6	3.6	0.3
	DBStd	8.58	5.66	5.36	4.10	3.42	3.67	2.81	2.69	2.45	2.78	3.71	3.69	4.73
	HDD10.0	1913	407	382	342	163	28	1	0	0	8	90	193	300
	HDD18.3	4186	665	615	601	409	227	105	31	43	157	333	443	558
	CDD10.0	831	0	0	0	4	61	152	263	235	101	16	0	0
	CDD18.3	63	0	0	0	0	2	6	35	19	0	0	0	0
	CDH23.3	277	0	0	0	0	12	29	161	74	1	0	0	0
	CDH26.7	39	0	0	0	0	1	3	24	11	0	0	0	0
Wind	WSAvg	3.5	3.7	3.5	3.5	3.5	3.3	3.4	3.1	3.0	3.2	3.6	3.9	4.1
Precipitation	PrecAvg	627	52	40	33	31	34	55	64	77	56	68	63	54
	PrecMax	912	113	92	63	65	68	95	142	167	123	135	111	139
	PrecMin	332	9	5	3	5	6	15	3	6	6	9	1	13
	PrecStd	119	24	20	17	18	17	23	30	37	27	31	29	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	4.6	10.0	17.0	24.8	26.6	29.3	27.8	21.0	15.7	10.1	8.6
		MCWB	5.1	3.0	5.2	10.8	16.4	18.2	20.8	20.9	17.0	13.4	9.1	7.4
	2%	DB	4.7	3.2	6.6	14.2	21.8	23.5	26.7	25.2	19.5	14.2	9.2	7.3
		MCWB	4.2	2.4	3.9	8.7	15.3	17.8	20.7	19.6	16.0	12.7	8.3	6.5
	5%	DB	3.6	2.4	4.5	12.1	19.4	21.6	25.0	23.4	18.3	13.4	8.6	6.5
		MCWB	2.9	1.8	2.8	7.9	14.4	16.7	20.1	18.8	15.3	12.1	7.8	5.8
	10%	DB	2.6	1.7	3.2	10.0	17.0	19.9	23.4	22.0	17.2	12.4	8.0	5.3
		MCWB	2.1	1.2	1.9	6.6	12.9	15.8	19.2	18.1	14.7	11.3	7.2	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	3.5	6.5	11.7	18.2	20.1	22.5	22.1	17.7	14.0	9.4	7.7
		MCDB	6.0	4.2	8.8	15.2	21.7	24.0	26.6	26.2	20.0	14.7	9.8	8.3
	2%	WB	4.3	2.6	4.1	9.7	16.5	18.7	21.5	20.7	16.6	13.1	8.6	6.7
		MCDB	4.6	3.0	6.0	13.1	20.4	22.5	25.4	24.0	18.7	14.0	9.1	7.1
	5%	WB	3.0	2.0	3.1	8.2	15.0	17.4	20.5	19.5	15.7	12.3	8.0	5.9
		MCDB	3.4	2.3	4.3	11.2	18.8	20.8	24.3	22.6	17.6	13.2	8.5	6.3
	10%	WB	2.1	1.4	2.2	7.0	13.2	16.2	19.5	18.5	15.0	11.5	7.2	4.7
		MCDB	2.5	1.7	3.0	9.4	16.5	19.3	23.0	21.3	16.8	12.3	7.8	5.2
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.4	5.5	6.1	6.7	6.1	6.2	6.4	5.5	4.5	3.1	3.3
		MCDBR	2.8	3.3	6.1	9.6	10.4	9.1	8.8	8.9	6.8	4.6	3.0	2.8
		MCWBR	2.7	2.8	4.3	5.8	5.7	4.9	4.6	4.9	4.4	3.5	2.8	2.7
	5% WB	MCDBR	2.7	3.1	5.5	8.1	9.2	7.8	7.7	7.5	5.8	4.0	3.0	2.6
		MCWBR	2.7	2.7	4.1	5.4	5.4	4.7	4.6	4.6	4.1	3.4	2.9	2.6
Clear-Sky Solar Irradiance	taub	0.287	0.314	0.337	0.374	0.368	0.367	0.396	0.390	0.358	0.340	0.324	0.283	
	taud	2.169	2.196	2.248	2.307	2.402	2.440	2.380	2.394	2.465	2.404	2.232	2.139	
	Ebn at Noon	524	697	798	830	865	870	831	808	780	675	473	397	
	Edh at Noon	49	77	97	109	106	103	108	99	79	63	47	36	
All-Sky Solar Radiation	RadAvg	0.29	0.86	2.40	4.03	5.68	5.99	5.59	4.40	2.75	1.24	0.36	0.20	
	RadStd	0.05	0.16	0.24	0.45	0.51	0.33	0.44	0.39	0.32	0.16	0.07	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page